

Education

Chongqing University

PH.D. STUDENT IN WIND ENGINEERING

Chongqing, China

Sep. 2024 - Present

Shijiazhuang Tiedao University

M.ENG. IN WIND ENGINEERING

Shijiazhuang, China

Sep. 2021 - Jun. 2024

- Thesis: Data-driven reduced-order modelling methods for unsteady aerodynamics and fluid flows of the flow around bluff bodies.

Hebei University of Science and Technology

B.ENG. IN CIVIL ENGINEERING

Shijiazhuang, China

Sep. 2017 - Jun. 2021

Research Interests

Core research interests: I focus on **computational fluid dynamics**, **bluff-body aerodynamics**, and **particle-laden flows**. I study vortex shedding, wake transition, turbulent transport, inertial-particle clustering, void formation, and deposition using high-fidelity simulation and Lagrangian particle tracking. I also develop **reduced-order and data-driven methods**, including dynamic mode decomposition and LSTM-based models, for flow-field reconstruction and unsteady aerodynamic prediction.

Core research question: How do coherent wake structures govern unsteady aerodynamic loads and the transport of inertial particles around bluff bodies?

Honors and Awards (Selected)

2023-2024 **Top Ten Academic Stars of Graduate Students**, Shijiazhuang Tiedao University. [\[Link\]](#)

2022-2023 **National Scholarship for Graduate Students**, Ministry of Education of China. [\[Link\]](#)

2021-2022 **Excellent Graduate Student**, Top 10% of students at STDU.

2023 **Excellent Paper Award**, 32nd National Conference on Structural Engineering. [\[Link\]](#)

2019-2020 **Third Prize**, Pinming Cup National BIM Competition.

2018-2019 **First-Class Academic Scholarship**, Top 5% of students at HEBUST.

Publications (*equal contribution; †corresponding author)

PREPRINTS

[P1] xx

Shijie Liu, xx.

Under review at *Physics of Fluids*, 2026. [\[Paper\]](#)

JOURNAL ARTICLES

[J5] **Effects of angle of attack and feature-preserving reduced-order models for canonical bridge deck wakes**

Shijie Liu, Yuexin Cao, Zejun Qin, Jian Zhao, Luming An, Peng Guo, Zhen Zhang, and Qingkuan Liu.

Applied Sciences, 15(23):12670, 2025. [\[Paper\]](#)

[J4] **Reconstruction and short-term prediction of wind pressure field on cylindrical coal sheds under undisturbed conditions using dynamic mode decomposition**

Qingkuan Liu, Jing Huo, Shijie Liu, Zejun Qin, Haohan Li, and Zhen Zhang.

Journal of Wind Engineering and Industrial Aerodynamics, 258:106017, 2025. [\[Paper\]](#)

[J3] **LSTM-based aerodynamic modeling for unsteady flows around structures**

Shijie Liu, Zhen Zhang, Xue Zhou, and Qingkuan Liu.

Wind and Structures, 38(2):147–160, 2024. [\[Paper\]](#)

[J2] **Bridge unsteady aerodynamic force prediction based on long short-term memory networks**

Qingkuan Liu, Shijie Liu, Zhen Zhang, Xue Zhou, and Hongmiao Jing.

China Journal of Highway and Transport, 36(8):56–65, 2023. [\[Paper\]](#)

[J1] **Unsteady flow-field analysis of flow around a square cylinder based on dynamic mode decomposition**

Shijie Liu, Zhen Zhang, and Qingkuan Liu.

Engineering Mechanics, 2023. [\[Paper\]](#)

[C3] DMD-based unsteady flow-field analysis around a square cylinder
Shijie Liu, Zhen Zhang, and Qingkuan Liu.
32nd National Conference on Structural Engineering, 2023. (Best Paper Award) [Paper]

[C2] LSTM-based aerodynamic coefficient prediction for two-dimensional cylinder flow
Shijie Liu, Bo Hu, Zhen Zhang, and Qingkuan Liu. [Paper]

[C1] Multi-scale LSTM prediction of wind pressure on coal-shed surfaces
Jing Huo, Shijie Liu, Zhen Zhang, and Qingkuan Liu. [Paper]

Research Experience

Chongqing University	Chongqing, China
PH.D. STUDENT IN WIND ENGINEERING, ADVISED BY PROF. CRUZ Y. LI	Sep. 2024 - Present
• Research Topics: particle-laden bluff-body wakes, computational fluid dynamics, and Lagrangian particle tracking.	
Shijiazhuang Tiedao University	Shijiazhuang, China
M.S. IN WIND ENGINEERING, ADVISED BY PROF. QINGKUAN LIU AND PROF. ZHEN ZHANG	Sep. 2021 – Jun. 2024
• Research Topics: data-driven unsteady aerodynamics, dynamic mode decomposition, and reduced-order flow-field modeling.	

Academic Services

Teaching Assistant	BEE20203 Fluid Mechanics for Built Environment and Energy Engineering, Chongqing University; 58 lecture hours and four practical classes, Fall 2026. [Link]
Reviewer	Physics of Fluids.
Conference Volunteer	22nd National Conference on Structural Wind Engineering and 8th National Wind Engineering Graduate Forum, Chongqing, 2025. [Link]
Conference Volunteer	2nd Academic Conference on Extreme Loads and Protection of Bridge Structures, 2023. [Link]

Skills

Programming	Python, C++, LaTeX
GPU & ML	CUDA, PyTorch, TensorFlow
CFD Solvers	OpenFOAM, Fluent, STAR-CCM+, IAMR; iFlowX (in-house GPU-accelerated CFD solver)
Pre/Post-processing	ICEM CFD, Tecplot, Paraview
Wind Tunnel Experiments	Boundary-layer wind-tunnel testing, pressure measurements, force measurements, vibration measurements
Languages	Chinese, English; CET-4: 532, CET-6: 513

Invited Talks

[T1] XX
XX, XX, 20XX. [Link]

Press Coverage (Selected)

[M1] “Selected as One of Shijiazhuang Tiedao University’s Top Ten Academic Stars”
by New Media Center, Shijiazhuang Tiedao University, April 26, 2024. [Link]

[M2] “Awarded the National Graduate Scholarship”
by Wind Tunnel Laboratory, Shijiazhuang Tiedao University, October 31, 2023. [Link]

[M3] “Awarded Hebei Provincial Graduate Innovation Funding”
by Wind Tunnel Laboratory, Shijiazhuang Tiedao University, January 3, 2023. [Link]